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AI-Powered Driver Fatigue and Distraction Detection Cameras

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Abstract

Driver fatigue and distraction are critical contributors to road traffic incidents, particularly in high-risk industries such as oil and gas. This paper presents ADNOC's implementation of AI-powered Active Fatigue and Distraction Detection (AFDD) cameras across its operational and contractor fleets to proactively mitigate fatigue- and distraction-related risks.

The AFDD system utilizes real-time artificial intelligence and computer vision to detect early indicators of driver fatigue and distraction, such as drowsiness, prolonged eye closure, mobile phone use, and loss of focus. Upon detection, the system delivers in-cabin alerts to prompt immediate driver correction. The deployment included device calibration, driver engagement programs, policy integration, and seamless interoperability with ADNOC's In-Vehicle Monitoring System (IVMS). A structured change management approach supported adoption and ensured organizational readiness.

Preliminary results show a measurable improvement in driver attentiveness, rest compliance, and proactive safety interventions. Early data suggests that the integration of AFDD technology significantly enhances situational awareness and supervisory oversight, supporting a broader safety culture transformation. This paper provides technical and operational insights into the design, integration, and early outcomes of an AI-based fatigue monitoring system in one of the world's largest oil and gas transportation networks.

Through detailed documentation of implementation challenges, change management, and system performance, this work offers additive value to existing literature. It provides practical, scalable lessons for petroleum industry stakeholders aiming to harness AI for behavioral risk mitigation in transport operations.

The initiative supports the United Nations Sustainable Development Goal (SDG) 3.6, which aims to halve the number of global road traffic deaths and injuries by 2030, and aligns with land transportation safety recommendations by the International Association of Oil & Gas Producers (IOGP).

Introduction

The oil and gas industry operates within a high-risk ecosystem, where the movement of people, equipment, and hazardous materials occurs daily across vast and often challenging terrains. Land transportation, in particular, remains one of the most vulnerable operational activities, consistently ranking among the top contributors to serious and fatal incidents within the industry. Despite continuous improvements in vehicle

safety and driver training, human error especially in the form of fatigue and distraction, continues to account for a significant proportion of transportation-related accidents.

Driver fatigue is a subtle but dangerous condition that impairs reaction times, reduces situational awareness, and increases the likelihood of critical decision-making errors. Distraction, whether due to mobile device usage, external stimuli, or cognitive overload, can be equally hazardous. Studies by the World Health Organization (WHO) and the International Association of Oil & Gas Producers (IOGP) have consistently highlighted fatigue and distraction as key behavioral risk factors behind roadway crashes, particularly in operations requiring long-distance driving, night shifts, and repetitive routes.

In response to this ongoing challenge, ADNOC "one of the world's leading energy producers" has prioritized the integration of advanced technologies to transform its transportation safety strategy. A cornerstone of this transformation is the deployment of AI-powered Active Fatigue and Distraction Detection (AFDD) Cameras, designed to proactively identify risky driver behavior before it results in a serious incident.

The AFDD system uses artificial intelligence and machine vision to continuously analyze a driver's facial and ocular behavior in real time. The technology is capable of detecting signs of drowsiness, yawning, prolonged eye closure, gaze deviation, and mobile phone usage. Upon detecting such behavior, the system provides immediate feedback to the driver via audible alerts. Simultaneously, event data is transmitted to ADNOC's centralized safety platform for logging, analysis, and corrective follow-up.

This paper documents ADNOC's end-to-end journey in piloting and scaling the AFDD system across both its corporate fleet and contractor vehicles. It outlines the motivation behind the initiative, the technical capabilities of the deployed solution, and the methodology used to integrate it within the broader transport safety ecosystem. In addition, it shares key insights gained during system calibration, user training, stakeholder engagement, and behavioral data analysis.

The paper further evaluates early performance metrics and user feedback gathered during the initial rollout phase. While the full impact of the technology on accident rates and long-term driver behavior will be assessed over time, preliminary results suggest a positive trend in terms of heightened driver alertness and reduced instances of non-compliance with rest guidelines.

Finally, the work explores how the AFDD initiative contributes to ADNOC's broader HSE strategy and its alignment with international safety benchmarks such as IOGP 365 and SDG 3.6. It offers industry peers a replicable model for leveraging AI-based driver monitoring technologies in complex, high-risk logistics operations.

By shedding light on both the technical and human dimensions of implementing AI in driver safety, this paper aims to stimulate cross-sector dialogue and encourage broader adoption of behavioral analytics in transport safety programs globally.

Statement of Theory and Definitions

The transportation of personnel and goods in the oil and gas industry involves numerous risks among them, the growing threat of driver fatigue and distraction. Historically, ADNOC and other energy-sector leaders have invested heavily in road safety through programs targeting vehicle compliance, driver qualification, journey management, and incident investigation. Yet, despite these efforts, human behavioral factors remain stubbornly difficult to control and predict.

As part of our continuous improvement, ADNOC conducted a comprehensive analysis of its land transportation incident data across both its own operations and those of its contractors. The findings echoed broader industry trends: a significant percentage of serious and near-miss road traffic incidents were linked to either fatigue or distraction. In many cases, these incidents occurred despite the presence of otherwise compliant vehicles, trained drivers, and approved journey plans highlighting a persistent blind spot in real-time behavioral risk monitoring.

Fatigue-related incidents tend to manifest in gradual degradation of attention and response. Drivers experiencing micro-sleep episodes or cognitive fatigue often fail to recognize their deteriorating condition until it's too late. Meanwhile, distraction events ranging from mobile phone use to off-road gaze or mental disengagement, can result in sudden and unpredictable driving errors. Such conditions are difficult to detect using traditional safety tools, including standard IVMS systems, which are more effective at flagging mechanical violations (e.g., over speeding, harsh braking) than internal states of the driver.

Recognizing this challenge, ADNOC began exploring next-generation technologies capable of detecting risk from the source which is the driver's own behavior in real time. Benchmarking studies were conducted against peer companies in the oil and gas sector as well as in logistics-intensive industries like aviation, mining, and public transportation. A recurring theme emerged: the increasing deployment of AI-enhanced vision systems to monitor driver fatigue and distraction.

These systems use infrared and visible-light cameras trained on the driver's face, in combination with AI models that interpret eye movement, blink rate, gaze direction, yawning frequency, and head pose. Unlike manual observation or retrospective data analysis, these tools offer the advantage of continuous, objective, and real-time monitoring, providing timely interventions before an incident can occur.

In early 2024, ADNOC initiated a pilot study to evaluate the technical readiness, user acceptance, and operational value of such systems in desert and urban driving conditions. The study was aligned with ADNOC's strategic HSE vision to advance predictive safety capabilities across its high-risk operations. It also supported the Group's broader sustainability goals under SDG 3.6, aimed at reducing road traffic fatalities and injuries by 50% by 2030.

The pilot yielded promising early outcomes, both in terms of driver compliance with safety protocols and improved supervisory awareness of behavioral trends. It also highlighted areas that would need to be addressed for full deployment:

- System calibration across different cabin types and lighting conditions
- Ensuring data integration with existing ADNOC IVMS and reporting platforms
- Managing privacy and data sensitivity concerns
- Establishing response protocols for AFDD-generated alerts
- Training and change management for driver buy-in

These challenges informed the design of a phased rollout strategy, wherein learnings from the pilot were embedded into a standardized implementation of the new Transportation Safety Corporate Practice. This Corporate Practice aimed not only to deploy hardware and software at scale, but also to embed a new layer of behavioral accountability and visibility into ADNOC's transportation safety ecosystem.

The decision to expand the AFDD deployment across ADNOC's fleet and soon to its contractors, was backed by a growing consensus among internal stakeholders that proactive behavioral monitoring represents the future of road safety in high-risk industries. With the convergence of AI, telematics, and behavioral science, ADNOC saw the opportunity to lead a transformation in how human risk is managed in Transportation.

Description and Application of Equipment and Processes

Aligned with IOGP 365 the Active Fatigue and Distraction Detection (AFDD) system deployed by ADNOC is designed to provide continuous, real-time monitoring of driver alertness using an integrated suite of hardware and software components. The system was selected and configured to meet the operational demands of land transportation in both blacktop and desert environments, and to align with international land transportation safety practices.

System Components

The AFDD platform consists of the following main components:

Driver-Facing Camera Unit. Positioned within the vehicle cabin, this unit features a wide-angle lens to capture the full facial profile of the driver regardless of seating position. The camera is equipped with infrared capabilities, enabling performance in both daylight and low-light conditions. An embedded image processor allows for edge computing, reducing latency in detection and alerting.

AI-Powered Detection Algorithms. The system uses pre-trained machine learning models to analyze driver facial behavior and detect indicators of fatigue or distraction. These include prolonged eye closure, reduced blink rate, gaze deviation, head drooping, yawning, and mobile phone use near the face. The algorithms continuously assess these indicators to generate real-time alerts when predefined risk thresholds are exceeded.

In-Vehicle Alert Interface. Upon detection of unsafe behavior, the system triggers an immediate in-cabin alert. Alerts include auditory tones or voice prompts that notify the driver and prompt corrective action. Alert latency is typically less than one second from detection to activation.

Telematics Integration Module. The AFDD unit is integrated with the vehicle's existing In-Vehicle Monitoring System (IVMS), allowing behavioral alerts to be logged alongside traditional telemetry data such as speed, seatbelt status, harsh braking, and GPS location. All events are transmitted in real time through secure 4G/LTE connections to ADNOC's centralized safety platform.

Central Monitoring and Analytics Platform. Collected data is stored and visualized on a web-based dashboard, enabling HSE teams and transport supervisors to review trends, generate reports, and identify high- risk routes, shifts, or driver profiles. Behavioral data is time-stamped and linked to specific journeys, creating a complete view of transport safety performance.

Alignment with IOGP Guidelines

ADNOC's AFDD deployment was built in strict accordance with IOGP 365 guidance, ensuring the system met or exceeded all relevant safety and technical requirements. Key areas of alignment include:

Table 1—ADNOC AFDD Implementation Against IOGP 365 Requirement

IOGP Requirement	AFDD Implementation at ADNOC
Continuous monitoring of driver state	100% of trips actively monitored in real time
Timely driver alerts	Immediate In-Vehicle alert of unsafe behavior
Overall Integration with performance management systems	AFDD data feeds integrated into IVMS platform
Post-event analysis and reporting	HSE Cockpit centralized dashboards and events tagging for review
Use in both day and night driving conditions	IR-capable cameras certified for 24 hr. operations
Driver privacy and ethical usage	Video not retained unless incident escalates

This alignment was essential to secure cross-functional approval for the system's deployment, ensuring not only technical robustness but also internal stakeholder confidence.

Field Application and Calibration

To ensure operational effectiveness, the AFDD system was field-tested across various vehicle types, including light vehicles, buses, and heavy transport trucks. Calibration activities included:

- Testing across multiple cabin configurations and lighting conditions

- Fine-tuning detection thresholds to minimize false positives
- Validating infrared performance under desert glare and nighttime operations

The system was adjusted to account for different driver postures, seat positions, and equipment layouts to ensure consistent performance across the ADNOC fleet.

Structured Rollout Strategy

ADNOC developed a phased rollout strategy, ensuring scalability and minimal disruption to daily operations. The full-scale implementation was approached:

- Site and Vehicle Readiness: vehicles were categorized based on operational profiles. Special attention was given to:
 - All buses and means of mass transportation
 - All fleet vehicles which are utilized in fatigue induced journeys (Journeys exceeding 90 minutes in duration)
 - All trucks and vehicles utilized to transport dangerous goods.
 - Any other vehicles that are determined based on GC/ Entity internal site- specific risk assessment & subject to ALARP criteria
- Driver Training and Engagement: Comprehensive training sessions were organized for drivers prior to the system going live. These sessions covered:
 - An overview of the AFDD system and its role in enhancing safety
 - Interpretation of alert signals and appropriate response protocols
 - Data privacy considerations and the benefits of real-time behavioral monitoring
- Feedback and Continuous Improvement: regular review sessions were instituted to collect feedback and make iterative improvements. This feedback loop was crucial for adjusting system parameters, especially in relation to alert sensitivity and contextual integration with route and shift data.

Change Management and Stakeholder Engagement

A critical element of the deployment methodology was an effective change management strategy. Recognizing that behavioral monitoring systems often prompt apprehension among drivers, ADNOC implemented a comprehensive communication plan designed to:

- Educate and Inform: Clear and consistent messaging was disseminated through multiple channels, including pre-rollout briefings, digital newsletters, and interactive workshops. The communication stressed that the AFDD system was implemented to enhance safety, not to penalize drivers.
- Collaborative Governance: A series of workshops and awareness campaigns were launched to drive the rollout process, creating spaces where concerns could be addressed in real time and robust support structures established to guide teams through each stage of the transition.
- Feedback Mechanisms: Anonymous feedback channels allowed drivers to report issues and suggest improvements. This ongoing dialogue helped refine both the system's technical performance and its operational integration.

Presentation of Data and Results

The initial deployment phase of the Active Fatigue and Distraction Detection (AFDD) system across ADNOC's land transportation fleet captured a substantial volume of behavioral data, offering insight into

system performance, driver response, and early safety improvements. The results presented below are drawn from the first four (4) months of operational use.

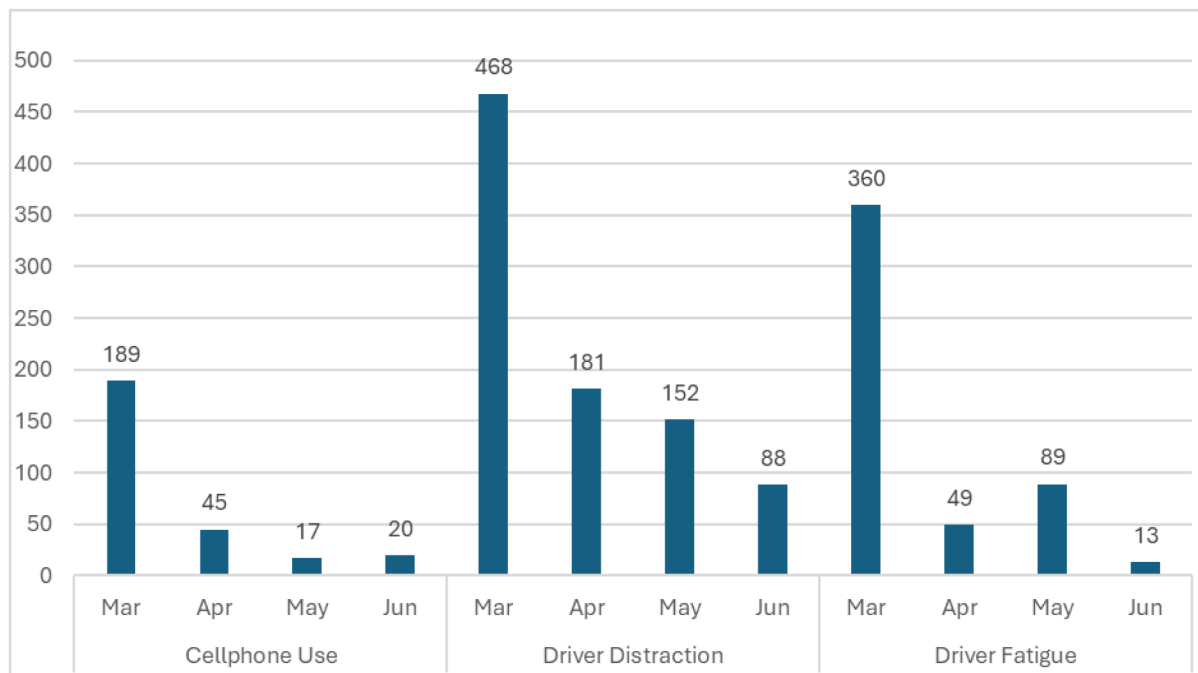


Figure 1—ADNOC AFDD recorded Events (Cellphone Use, Driver Distraction, and Driver Fatigue) showing a consistent downward trend over the past four months which shows the effectiveness of the AFDD implementation

System Deployment and Operational Scope

During the monitoring period, the AFDD system was installed in 370 vehicles spanning light vehicles, heavy trucks, and buses. The combined distance driven exceeded 4.5 million Km across desert and blacktop conditions. The system operated continuously, providing 24/7 behavioral monitoring during all active trips.

Detected Behavioral Events

The AFDD system identified multiple categories of fatigue- and distraction-related behavior. Key indicators tracked included:

- Cell phone use
- Driver Distraction
- Driver Fatigue

Each of these behaviors was detected in real time and classified based on severity. Data was aggregated by trip, time of day, route type, and driver profile.

Impact on Safety Performance

Although the AFDD system had not been in place long enough to produce statistically significant reductions in crash rates across the entire fleet, early indicators showed a positive trend. A confirmed 12 number of fatigue incidents were prevented as a direct result of having the AFDD in the vehicle.

Additionally, fatigue-related incidents that were previously difficult to quantify (and often underreported) could now be systematically tracked, giving supervisors a clearer picture of potential risk concentrations.

Strategic Implications for ADNOC and the Oil & Gas Industry

The AFDD system reinforces ADNOC's position as a regional and global leader in HSE performance. By integrating cutting-edge AI technology directly into the operational workflow of land transportation, ADNOC has taken a definitive step toward data-driven safety leadership.

For the oil and gas sector, where transport-related fatalities consistently rank among the top contributors to workforce harm, this represents a blueprint for transformation.

ADNOC's model demonstrates how safety management systems can be enhanced through digital solutions that go beyond compliance and actively shape behavior in real time.

Furthermore, the deployment of AFDD supports several strategic goals simultaneously:

- Operational Risk Management: By providing real-time visibility into driver state, AFDD allows for earlier intervention, reducing the potential for high-impact transport incidents.
- Sustainability Alignment: The initiative supports ADNOC's commitment to SDG 3.6 and IOGP 365, reinforcing its contribution to global road safety efforts.
- Technology Leadership: Implementing and integrating behavioral AI at scale sets a precedent for other industrial operators exploring similar applications.

Challenges and Mitigation Strategies

Implementing AFDD technology across such a large and diverse fleet was not without challenges. A few key areas of concern emerged:

Human Factors Resistance

Some drivers expressed discomfort at being continuously monitored, fearing punitive outcomes or misuse of video data. To mitigate this, ADNOC implemented a transparent communication strategy, emphasizing the system's safety purpose, enforcing privacy safeguards, and positioning AFDD data as a coaching not punishment tool.

System Calibration Variability

Different vehicle cabin designs, driver seating postures, and environmental lighting created early challenges in maintaining consistent algorithm performance. These were resolved through:

- Tailored calibration profiles for different vehicle classes
- Infrared tuning for desert glare and low-light conditions
- Automated update rollouts to enhance model accuracy over time

Data Management Complexity

Integrating high-frequency behavioral data with existing telematics platforms and ensuring clean, contextual data streams required advanced data engineering. ADNOC's IT and HSE teams collaborated to design a layered architecture that supports:

- Real-time alert handling
- Event-driven analytics
- Scalable storage with privacy controls

Organizational Learning and Safety Culture Transformation

One of the most profound shifts observed during implementation was the change in organizational mindset. Historically, road safety conversations within ADNOC were focused on mechanical compliance (vehicle

inspection, speed, harsh braking). With the integration of AFDD, the conversation expanded to include the mental and behavioral state of drivers, bringing attention to often overlooked precursors of incidents.

This shift has enabled ADNOC to:

- Transition from incident-based analysis to leading indicator monitoring
- Identify behavioral patterns that may not trigger IVMS violations but still carry risk
- Drive a coaching culture, where supervisors engage drivers based on real-time performance data
- Foster cross-departmental collaboration between operations, safety, IT, and training units

System Reliability and Limitations

During the rollout, the AFDD system was always functioning, with minimal technical issues. However, some challenges were identified:

- False positives in distraction detection when drivers turned their heads to check for oncoming traffic
- Reduced accuracy in cases where drivers wore polaroid sunglasses
- Privacy concerns raised by a minority of drivers, especially around long-term data retention (addressed via robust governance measures and anonymization protocols)

Continuous algorithm improvement cycles enabled via cloud-based updates, helped to address many of these limitations over time.

Conclusions

The introduction and deployment of AI-powered Advanced Fatigue and Distraction

Detection (AFDD) technology across ADNOC's transportation fleet represents a pivotal advancement in proactive road safety management. In an operational context where traditional safety systems often struggle to identify behavioral risks in real time, the AFDD system has filled a critical gap offering a scalable, intelligent, and integrated solution to monitor, analyze, and influence driver behavior.

Behavioral Risk is Measurable and Controllable

Through continuous facial and attentional monitoring, ADNOC has demonstrated that it is possible to detect, quantify, and reduce driver fatigue and distraction with high reliability. Real-time alerts serve as effective interventions, while post-event analytics enable structured follow-up and performance improvement.

AI Can Serve as an Enabler of Human-Centric Safety

Contrary to concerns about surveillance or automation replacing human oversight, the AFDD system has empowered drivers and supervisors with actionable, data-informed insights. It complements existing IVMS system by adding visibility into human factors previously missed by mechanical telemetry.

Safety Culture is Reinforced, Not Replaced, by Technology

The rollout of AFDD has catalyzed a deeper organizational shift—from reactive incident analysis to proactive safety coaching. The program's success depended not only on technical robustness but also on transparent communication, training, and alignment with ADNOC's values.

Data-Driven Decisions Improve Predictive Capabilities

The aggregation and analysis of fatigue-related behavior across time, geography, and driver cohorts have helped ADNOC identify systemic risks, tailor coaching interventions, and plan safer journeys. These data insights are now embedded into monthly safety reviews and performance dashboards.

IOGP-Aligned Solutions Can Scale Across Diverse Fleets

The adoption of IOGP 365 aligned system architecture allowed ADNOC to deploy the AFDD platform across different vehicle types, terrains, and operating companies while maintaining consistency in reporting and compliance monitoring.

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Nomenclature

Table 2—Definition and Abbreviations

Term / Symbol	Definition
AFDD	Active Fatigue and Distraction Detection – an AI-based in- vehicle system used to monitor driver behavior in real time.
IVMS	In-Vehicle Monitoring System – a telematics system that tracks vehicle movement, driver behavior, and operational compliance.
ALARP	As Low As Reasonably Practicable – a principle used to reduce risk to a level that is as low as is reasonably achievable.
SDG 3.6	Sustainable Development Goal 3.6 – a United Nations target aiming to halve the number of global road traffic deaths and injuries by 2030.
AI	Artificial Intelligence – computer systems capable of performing tasks that typically require human intelligence, such as pattern recognition and decision making.

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